DEVIATION REPORT - PHARMACEUTICAL QUALITY SYSTEM
Document Ref: DR-2026-0842 | Unit: API Synthesis Building 4

Title: Reactor Temperature Excursion During Synthesis Step 3
Date of Occurrence: 14-07-2024
Time of Occurrence: 09:15 IST
Department / Source: Manufacturing
Product: Ibuprofen Active Pharmaceutical Ingredient
Batch Number: API-2407-RX
Batch Size: 48 kg

DETAILED DESCRIPTION:
During the synthesis of Ibuprofen batch API-2407-RX on 14-Jul-2024, the jacketed reactor R-102 temperature rose to 92°C against an approved operating range of 70°C to 80°C. The excursion lasted approximately 25 minutes before the operator noticed and manually adjusted the cooling water bypass flow. Approximately 48 kg of in-process material was affected in the vessel. No visible discoloration or precipitation was observed at the time. The batch has been isolated and placed on physical QA quarantine hold pending critical review.

INITIAL ROOT CAUSE HYPOTHESIS:
Cooling water supply control valve V-204 experienced an actuator signal failure due to compressed air pressure fluctuation. Facilities engineering has been dispatched to inspect pneumatic lines.

SEVERITY CLASSIFICATION:
Severity: Major
Impact Category: Product Quality

IMMEDIATE CONTAINMENT TAKEN:
1. Synthesis reaction paused and brought back to 72°C.
2. In-process batch quarantined and tamper-evident tags applied.
3. Quality Assurance and Shift Lead notified immediately.

Reported By: M. Rajan (Senior Process Operator)
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